

University of Dhaka
CSE 2106: Microprocessor and Assembly Language Lab
Date: 05.11.2025

1. Write an assembly program that calls a function **remove_first** that takes an array and its length and removes the first element of the array.

Input:

R0 = array base address, R1 = length

2. Write an assembly program that calls a function **remove_last** that takes an array and its length and removes the last element of a list.

Inputs:

R0 = array base address, R1 = length

3. Write an assembly program that calls a function **sum_avg** that takes an array and its length, returns the **sum** of all elements in R3, and the **average** in R4.

Input:

R0 = array base address, R1 = length

4. Write an assembly program that calls a function **find_min_max** that takes an array and its length and returns minimum in R3, maximum in R4

Input:

R0 = array base address, R1 = length

5. Write an assembly program that calls a function, **reverse_array, which takes an array and its length, and reverses** a 32-bit integer array in memory. $\text{arr}[] = \{1, 2, 3, 4\} \rightarrow \text{arr}[] = \{4, 3, 2, 1\}$

Input:

R0 = array base address, R1 = length

6. Write an assembly program that calls a function **odd_even** that takes an array and its length, and counts how many even and odd numbers exist in a given array.

Output:

R0 = number of even numbers, R1 = number of odd numbers

7. Write an assembly program that calls a function **rotate_k** that takes an array, its length, and k (rotation offset), and rotates the array left by k positions.

Input:

R0 = base, R1 = length, R2 = K (rotation offset)

Example:

arr = [1, 2, 3, 4, 5], k = 2 $\rightarrow$ [3, 4, 5, 1, 2]